CH-PROJ-M · T6 · UNI BAMBERG · SS 2026
Computational Humanities — Master's Project

2026 · 05 · 18
5 min talk · 5 min Q&A

PROJECT

Audio-to-Text Captioning with **Large Audio-Language Models**

Can these models describe sounds that **overlap***?*

P2 — MILESTONE 2 OF 5

Data Acquisition Strategy

- 01 Test corpus · Clotho v2.1
- 02 Acquisition · which mics?
- 03 Properties · + annotation
- 04 Classes · polyphony test
- 05 Risks · path forward

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P1 CALLBACK
Closing the loop on **contamination**
— the Risk to Watch I named last time.

P2 · DATA ACQUISITION STRATEGY

Clotho v2.1 — and why it is the right test corpus

Zero-shot evaluation, so no recording. The question is which test corpus.

CHOSEN — CLOTHO V2.1

- ✓ **DCASE Task 6 official benchmark** — leaderboard parity
- ✓ **5 captions per clip** — dense annotation enables under-description detection
- ✓ **FreeSound-sourced**, not AudioSet — lower contamination risk than AudioCaps

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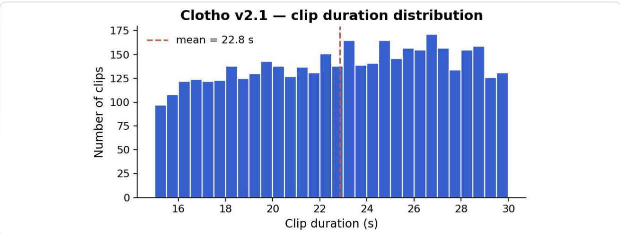
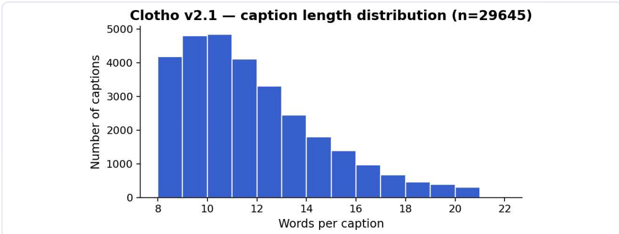




CONSIDERED & REJECTED

- ✗ **AudioCaps** — derived from AudioSet (LALM training overlap)
- ✗ **AudioSetCaps** — same AudioSet family
- ✗ **WavCaps** — web-scraped, overlap risk with Clotho / AudioCaps
- ✗ **Cacophony** — contrastive training corpus, not an eval benchmark

FRAME *The ALM Datasets Survey 2025 calls unverified pretrain overlap the* **"zero-shot illusion."** *This deck makes our test-corpus choice defensible against it.*

ACQUISITION

Which mics? — community-curated, heterogeneous devices

The honest answer is all of them. Device mismatch is built in.

Curated, not recorded.

Drossos, Lipping & Virtanen (Tampere) assembled Clotho from **FreeSound** uploads. The team's contribution is the captions — every clip carries its uploader's recording device.

Device mismatch is built in.

Studio condensers, field recorders, mobile phones with MEMS mics — Clotho samples **all of them**. This is a generalisation hazard we acknowledge; we treat it as a realistic property of deployment audio.

The strip below makes it visible.

Same nominal sound (rain) from three corpora. Length, SNR & recording quality differ visibly across Clotho / AudioCaps / WavCaps.

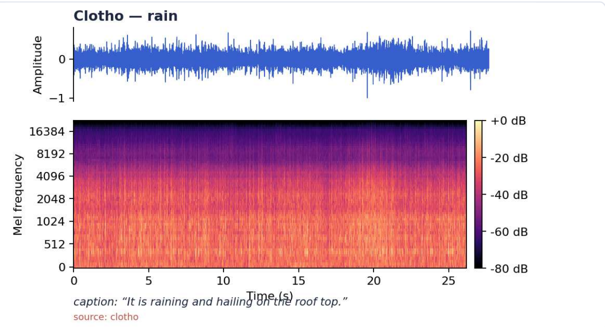
DEVICE MISMATCH · IN ONE LINE

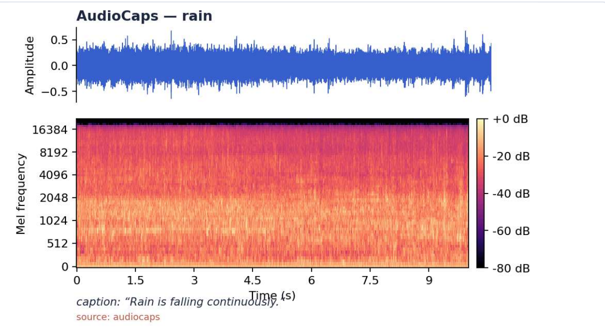
Studio condenser vs. **MEMS mic**

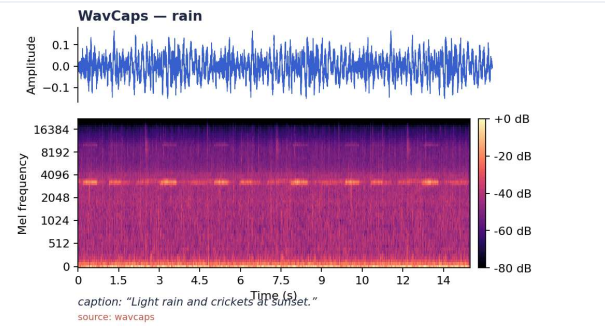
Frequency response, self-noise & SPL handling differ by an order of magnitude. **Real deployment audio is a mixture of both** — Clotho's heterogeneous FreeSound provenance is a feature, not a bug.

- Studio condenser
- MEMS phone mic

SAME NOMINAL SOUND — RAIN — THREE CORPORA







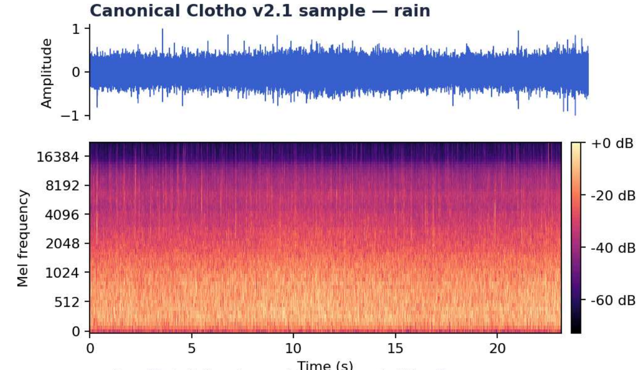
PROPERTIES & ANNOTATION

Audio properties + annotation

Weak-label benchmark, 5 captions per clip, spectrograms via `librosa.stft`.

AT A GLANCE

Sample rate	**44.1 kHz**
Channels	**1 (mono)**
Bit depth	**16-bit**
Format	**WAV (PCM)**
Clip duration	**15 – 30 s**
Captions / clip	**5**
Caption length	**8 – 20 words**
Vocabulary	**open (no fixed taxonomy)**
Licence	**CC-BY 4.0**
Distribution	**Zenodo record 4783391 · FAIR**



caption: "Rain falling down onto a house or building."

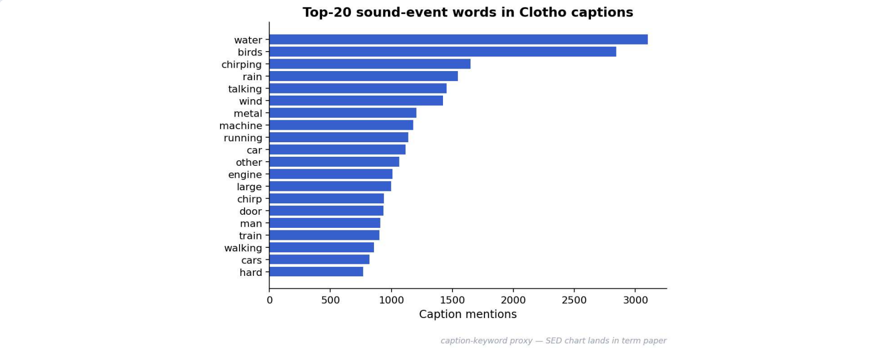
ANNOTATION TYPE · WEAK (CLIP-LEVEL, NO TIME-STAMPS)

5 human-written captions per clip, 8–20 words, crowdsourced via Amazon MTurk. Post-processed to remove unique words, named entities & speech transcription — **Clotho is not a speech corpus.**

Strong (segment-level) labels are **derived later** via PaSST / PANNs SED for the polyphony split.

CLASSES · EXAMPLES · THE FORMAL TEST

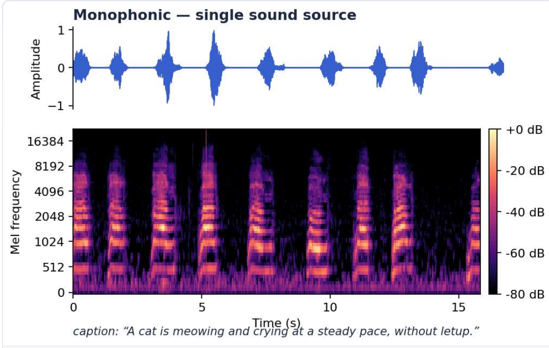
Sound classes, examples — and how I detect under-description

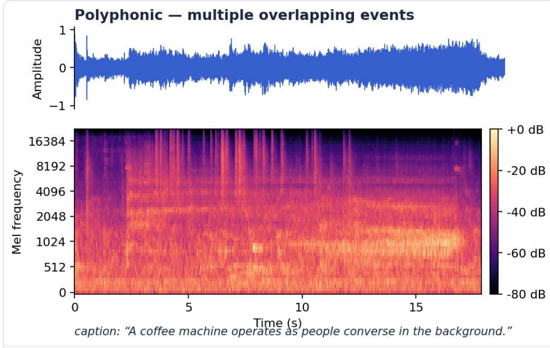


→ **Open-vocabulary** captions — no fixed taxonomy. Pseudo-classes derived by PANNs tagging.

→ **Long-tail distribution** dominated by *music*, *water*, *traffic*.

POLYPHONY BUCKET · SINGLE-SOURCE VS. MIXED-SOURCE





RQ2 · UNDER-DESCRIPTION TEST

A clip is under-described iff

$$\mathrm{card}(E_{model}) < \mathrm{card}(E_{ref} \cap E_{audio})$$

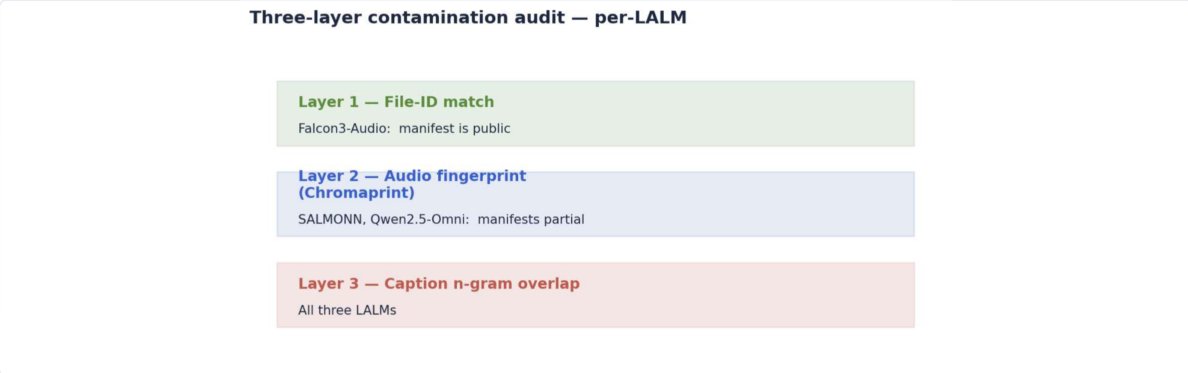
E_{audio} = SED-detected events · E_{ref} = union of 5 captions · E_{model} = model output entities.

Δ MACE = poly-bucket rate − mono-bucket rate.

RISKS · PATH TO EVALUATION

Risks I am closing on this slide

*P1 closed with **contamination** as my Risk to Watch and an open question: do the three LALMs fail polyphony the **same way**, or differently? Both shape this slide.*



EMPIRICAL WEDGE

FD-DeCap (IEEE TASLP 2025, doc 11333308) reports SPIDEr after causal-inference debiasing:

0.282 CLOTHO vs. **0.429** AUDIOCAPS

The contaminated corpus scores **~50 % higher** — the direct empirical wedge for the Clotho choice.

- **Audit scope:** the 3 LALMs only (Falcon3-Audio · SALMONN · Qwen2.5-Omni). The 3 traditional baselines (CNN14 / AST / EnCLAP) use small, fully-known corpora — no

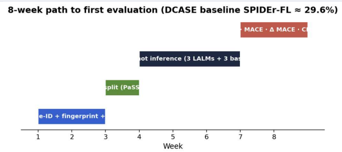
HOW CLOTHO SERVES EACH RESEARCH QUESTION

Out of scope: original recording (zero-shot protocol).

RQ1 Baseline parity
SPIDEr-FL · CIDEr
DCASE Task 6 leaderboard parity — direct comparison to the 29.6 % CNN14 baseline.

RQ2 Polyphony · core
Δ MACE (poly − mono)
5-caption union = E_{ref}; PaSST / PANNs SED on Clotho dev = E_{audio}.

RQ3 Hallucination
CHAIR-audio · MACE Precision
5-caption density makes invented-entity detection statistically reliable.



REFERENCES

References — IEEE format

[1] Drossos, K., Lipping, S., Virtanen, T. "Clotho: An Audio Captioning Dataset." *IEEE ICASSP*, 2020. `Zenodo record 4783391 (v2.1).`

[2] Falconi et al. "Competitive Audio-Language Models with Data-Efficient Single-Stage Training on Public Data (Falcon3-Audio)." *IEEE ASRU*, 2025. `ieeexplore.ieee.org/document/11434596`

[3] Dixit et al. "MACE: Leveraging Audio for Evaluating Audio Captioning Systems." *IEEE ICASSP-W*, 2025. `ieeexplore.ieee.org/document/11011270`

[4] IEEE ASRU 2025. "Data Leakage Benchmark." `ieeexplore.ieee.org/document/11450559`

[5] Mei et al. "Beyond the Status Quo: Critical Reflection on Audio Captioning Metrics." *IEEE/ACM TASLP*, 2023. `doi:10.1109/TASLP.2023.3321968`

[6] Dixit / Khare et al. "FD-DeCap: A Front-Door Causal Inference-Based Framework for Debiasing Audio Captioning." *IEEE/ACM TASLP*, 2025. `ieeexplore.ieee.org/document/11333308`

[7] Kong et al. "PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition (CNN14)." *IEEE/ACM TASLP*, vol. 28, pp. 2880–2894, 2020. `doi:10.1109/TASLP.2020.3030497`

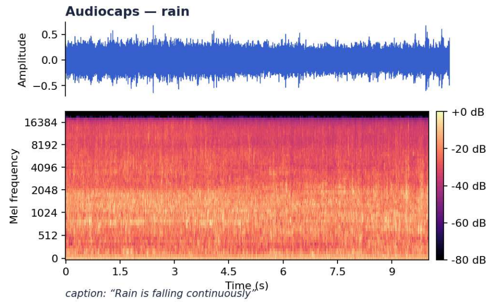
[8] Mei et al. "WavCaps: A ChatGPT-Assisted Weakly-Labelled Audio Captioning Dataset for Audio-Language Multimodal Research." *IEEE/ACM TASLP*, 2024. `arxiv.org/abs/2303.17395`

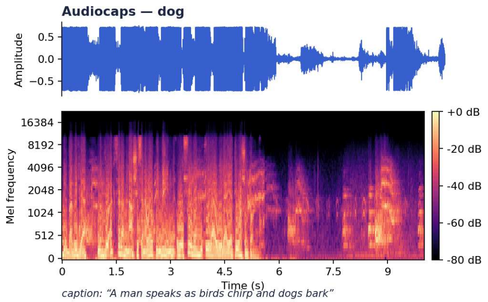
Shortlist of 8 data-strategy references from the project's 49-paper corpus. Full corpus + 38 read papers indexed in the project wiki.

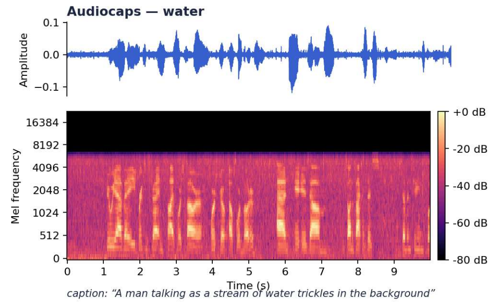
APPENDIX A · Q&A ONLY

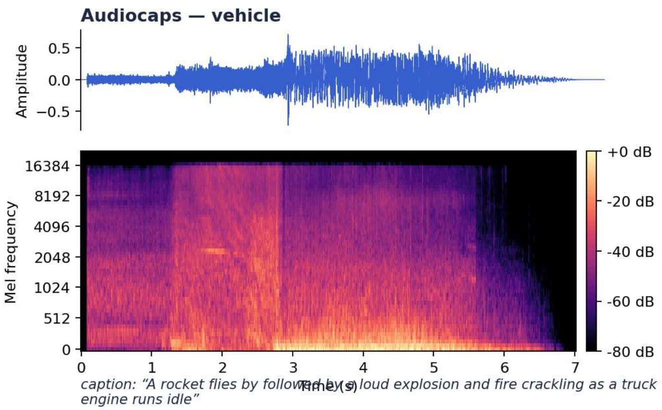
AudioCaps — five representative clips

Shown only if asked. AudioCaps = 10-second YouTube segments from **AudioSet**, *single-annotator captions. Note YouTube-encoding artifacts and uneven SNR.*

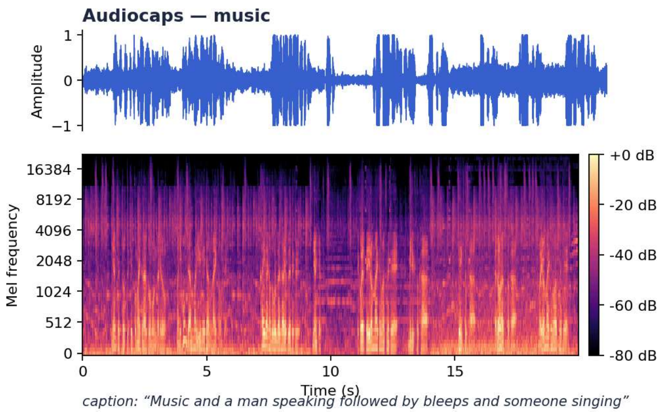








"A rocket flies by followed by a loud explosion and fire crackling as a truck engine runs idle."



"Music and a man speaking followed by bleeps and someone singing."

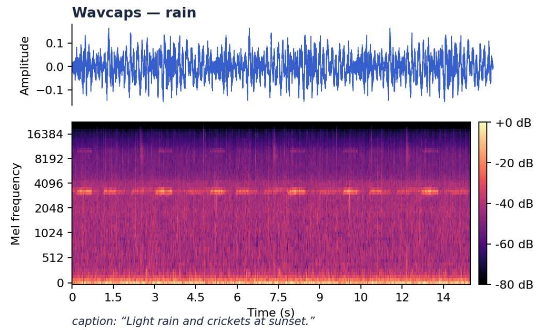
WHY THIS MATTERS

Captions are **single-annotator** — one of the three reasons **Clotho (5 captions per clip)** is preferred for a clean zero-shot benchmark.

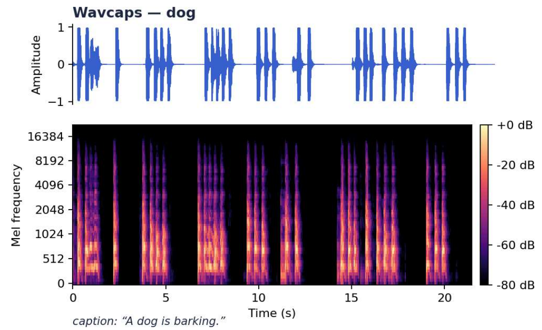
APPENDIX B · Q&A ONLY

WavCaps — SoundBible subset, six clips

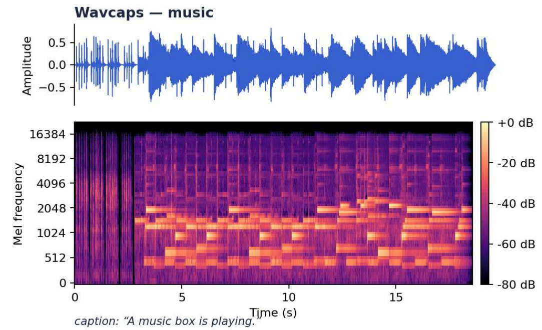
Shown only if asked. WavCaps = 400K-clip web-scraped corpus over four sub-sources (AudioSet_SL · BBC Sound Effects · FreeSound · SoundBible). SoundBible is the most curated subset.



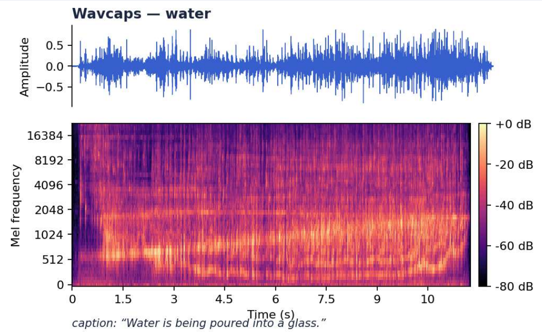
caption: "Light rain and crickets at sunset."



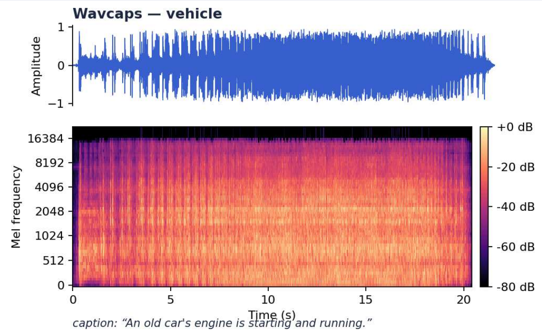
caption: "A dog is barking."



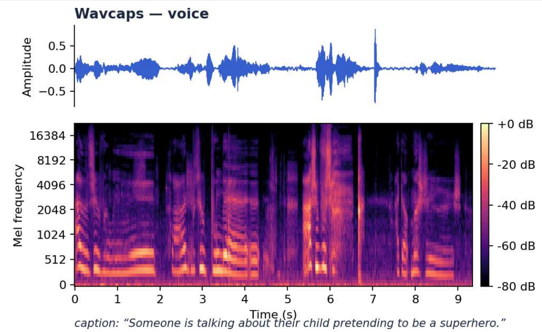
caption: "A music box is playing."



caption: "Water is being poured into a glass."



caption: "An old car's engine is starting and running."



caption: "Someone is talking about their child pretending to be a superhero."

WHY THIS MATTERS WavCaps captions are **LLM-generated from web metadata** — they inherit any biases of the generating LLM. The FreeSound subset overlaps Clotho's source platform — a contamination risk. Both are reasons Clotho is preferred for a clean zero-shot benchmark.